LIRUI WANG

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EDUCATION

University of Washington, Seattle, WA *Sep 2016 - Present*
Bachelor of Science, Electrical Engineering, (Honors), College of Engineering
Bachelor of Science, Computer Science, (Honors), College of Arts and Sciences
Bachelor of Science, Mathematics, (Minor), College of Arts and Sciences
Master of Science, Computer Science, College of Arts and Sciences
GPA: 3.96/4.00, *summa cum laude*, Advisor: Prof. Dieter Fox

RESEARCH PROJECTS

Deep Visuomotor Policies for Cluttered Scene Grasping *Jan 2020 - Present*

- Investigated deep RL methods for 6D grasping in cluttered scenes with expert experience.
- Compared different representations for robust generalization of imitation learning.

Joint Motion and Grasp Planning *March 2019 - Dec 2019*

- Implemented CHOMP and introduced an integrate planner that optimizes manipulation trajectory with online grasp synthesis and selection.
- Proposed an optimization-based approach and an online learning framework to generate and select grasp online.

Pose Tracking with the Deep Iterative Matching Network *June 2018 - March 2019*

- Evaluated DeepIM on YCB dataset and achieved robust results trained on synthetic data only.
- Improved network training stability, pose distribution and occlusion generation.
- Extended DeepIM to achieve state-of-the-art pose tracking accuracy and built real-world applications.
- Worked on a OpenGL renderer and kinematics tree for vision system application on robotics.

Unsupervised Flow and Depth Methods Study *March 2018 - June 2018*

- Investigated deep learning methods for 3D geometry tasks such as depths, flow, and egomotion.
- Explored unsupervised structures and ideas such as SfMlearner and MonoDepth and analyzed them on a manipulation setting.

Baxter Hand-Eye Coordination *Aug 2017 - March 2018*

- Used Baxter to grasp and manipulate daily objects detected by PoseCNN.
- Studied camera models, hand-eye calibration with depth camera, robot kinematics, 3D geometry and ROS packages for tracking and control.

WORK EXPERIENCE

Seattle, WA *June 2020 - Sep 2020*
Applied Research Intern, NVIDIA

Seattle, WA *Jan 2020 - Mar 2020*
Artificial Intelligence Teaching Assistant, UW CSE 473

Seattle, WA *Sep 2019 - Jan 2020*
Computer Vision Teaching Assistant, UW CSE 455

Shenzhen, China *Jun 2017 - Aug 2017*
Robotics Engineer Intern, DJI Technology

- Worked on autonomous robot battle which uses vehicles with TX1 onboard and DJI Manifold drones with Guidance.
- Implemented functions such as take-off, tag detection, and refilling to localize drone and communicate.
- Applied SLAM to mapping and localization, and built a state machine to generate plans.

LEADERSHIP AND INVOLVEMENT

Seattle, WA Sep 2016 - Feb 2018
Computer Vision Lead *Advanced Robotics at the University of Washington*

- Managed a team of 10 students to design computer vision algorithms for our robots in competition RoboMasters hosted by DJI.
- Performed system integration of controls, navigation and serial communication.

MISCELLANEOUS PROJECTS

- **Spatial Audio Processing in the SoundScape System**: Designed a sound processing pipeline from scratch for modelling the acoustics of a train.
- **Sheet Music Editor**: Created a demo app that can record sound pieces and translate into sheet music display UI using DTFT, and it was awarded 1st place out of 35 teams in Seattle.
- **Embedded Drowsiness Detection System**: Built an embedded system on Raspberry Pi that that simulates a real-time drowsiness detection system based on human face detection.
- **Survey and Empirical Evaluation of Attacks and Defense for Robust Classification**: Investigated example reweighting and adversarial training techniques to robustify a fossil classification task and surveyed recent papers on attacks and defense of adversarial examples for neural networks.
- **Continuous Trajectory Optimization with Machine Learning Techniques**: Summarized the formulation of trajectory optimization as functional gradient descent using Reproducing Kernel Hilbert Space and as probabilistic inference using Gaussian Process.
- **Microsoft Server Mover Project**: Built an autonomous mobile manipulator, a UR5 arm mounted on a Server Lift, to pull out servers from server rack and transport it to repair.

HONORS

- Annual Deans List, University of Washington, *2016 - 2020*
- Tau Beta Pi Scholarship *2017 - 2019*
- Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship *2017 - 2018*
- Patricia G. Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship *2018 - 2019*

PUBLICATIONS

- **Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection**
  Lirui Wang, Yu Xiang, Dieter Fox
  In Proceedings of Robotics: Science and Systems (RSS 2020)

COURSES

- **Computer Science**: Data Structure and Parallelism, Algorithms, Computer Vision, Artificial Intelligence, Deep Learning, Modern Algorithm Toolbox, Graphics, Robotics, Robustness in Machine Learning (Grad), Interactive Learning (Grad), Robotics (Grad), Reinforcement Learning
- **Electrical Engineering**: Digital Signal Processing and Filtering, Control Theory, Embedded System, Discrete Time Linear System, Systems, Controls, and Robotics Capstone, Optimization and Learning for Control (Grad)
- **Math**: Multivariable Calculus, Probability and Statistics, Numerical Analysis, Linear Analysis, Topology, Convex Optimization and Analysis (Grad), High Dimensional Statistics and Statistical Learning (Grad), Statistical Learning: Modeling, Prediction, And Computing (Grad)
- **English**: Technical Writing, Advanced Technical Writing in Electrical Engineering

SKILLS

- **Systems**: ROS, Ubuntu, ARM, Arduino, Nvidia TX Series, STM32
- **Languages**: Python, C++/C, Matlab, Java
- **Tools**: Git, CMake, OpenCV, Tensorflow, Pytorch, MXNet, OpenGL, CUDA, Qt, Bullet, VTK,
- **Hardware**: 3D Printing, Laser Cutting, Machining, Soldering